

Mathematics for Machine Learning

Vazgen Mikayelyan



The Weak Law of Large Numbers

The Weak Law of Large Numbers

Markov's theorem

Let X_n be a sequence of random variables, each having a finite mean ($\mathbb{E}[X_k] = a_k$) and variance. If

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \text{Var} \left(\sum_{k=1}^n X_k \right) = 0, \text{ then } \frac{1}{n} \sum_{k=1}^n (X_k - a_k) \xrightarrow{\mathbb{P}} 0.$$

The Weak Law of Large Numbers

Markov's theorem

Let X_n be a sequence of random variables, each having a finite mean ($\mathbb{E}[X_k] = a_k$) and variance. If

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \text{Var} \left(\sum_{k=1}^n X_k \right) = 0, \text{ then } \frac{1}{n} \sum_{k=1}^n (X_k - a_k) \xrightarrow{\mathbb{P}} 0.$$

Corollary

If X_n is a sequence of independent and identically distributed random variables, each having a finite mean ($\mathbb{E}[X_1] = a$) and variance, then

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{\mathbb{P}} a.$$

The Strong Law of Large Numbers

Borel's theorem

If X_n is a sequence of independent and identically distributed random variables, each having a finite moment of order 4 and $\mathbb{E}[X_1] = a$, then

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{a.s.} a.$$

The Strong Law of Large Numbers

Borel's theorem

If X_n is a sequence of independent and identically distributed random variables, each having a finite moment of order 4 and $\mathbb{E}[X_1] = a$, then

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{a.s.} a.$$

Kolmogorov's theorem

If X_n is a sequence of independent and identically distributed random variables, each having a finite mean ($\mathbb{E}[X_1] = a$), then

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{a.s.} a.$$

The Central Limit Theorem

Theorem

If X_n is a sequence of independent and identically distributed random variables, each having a finite mean μ and variance σ^2 , then the CDFs of

$$\frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}$$

tends to the standard normal distribution as $n \rightarrow \infty$.

Double Integrals

Double Integrals

Let $f : A \rightarrow \mathbb{R}$, where $A = [a, b] \times [c, d]$. Also let

$$P_1 = \{x_0, x_1, \dots, x_n\}, P_2 = \{y_0, y_1, \dots, y_n\}$$

are partitions of the segments $[a, b]$ and $[c, d]$ respectively.

Double Integrals

Let $f : A \rightarrow \mathbb{R}$, where $A = [a, b] \times [c, d]$. Also let

$$P_1 = \{x_0, x_1, \dots, x_n\}, P_2 = \{y_0, y_1, \dots, y_n\}$$

are partitions of the segments $[a, b]$ and $[c, d]$ respectively. Denote by λ the maximal length of the diagonals of the rectangles obtained by doing partitions.

Double Integrals

Let $f : A \rightarrow \mathbb{R}$, where $A = [a, b] \times [c, d]$. Also let

$$P_1 = \{x_0, x_1, \dots, x_n\}, P_2 = \{y_0, y_1, \dots, y_n\}$$

are partitions of the segments $[a, b]$ and $[c, d]$ respectively. Denote by λ the maximal length of the diagonals of the rectangles obtained by doing partitions.

Definition

The Riemann sum of a function $f(x, y)$ over this partition of A is

$$\sigma = \sum_{i=1}^n \sum_{j=1}^n f(u_i, v_j) \Delta x_i \Delta y_j,$$

where (u_i, v_j) is some point in the rectangle $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$ and $\Delta x_i = x_i - x_{i-1}$, $\Delta y_j = y_j - y_{j-1}$.

Double Integrals

Definition

The double integral of a function $f(x, y)$ in the rectangular region A is the following limit

$$\iint_A f(x, y) \, dx dy = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \sum_{j=1}^n f(u_i, v_j) \Delta x_i \Delta y_j$$

Double Integrals

Definition

The double integral of a function $f(x, y)$ in the rectangular region A is the following limit

$$\iint_A f(x, y) dx dy = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \sum_{j=1}^n f(u_i, v_j) \Delta x_i \Delta y_j$$

To define the double integral over a bounded region A other than a rectangle, we choose a rectangle B such that $A \subset B$ and define the function $g(x, y)$ so that

$$g(x, y) = \begin{cases} f(x, y), & \text{if } (x, y) \in A, \\ 0, & \text{if } (x, y) \in B \setminus A. \end{cases}$$

Double Integrals

Definition

The double integral of a function $f(x, y)$ in the rectangular region A is the following limit

$$\iint_A f(x, y) dx dy = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \sum_{j=1}^n f(u_i, v_j) \Delta x_i \Delta y_j$$

To define the double integral over a bounded region A other than a rectangle, we choose a rectangle B such that $A \subset B$ and define the function $g(x, y)$ so that

$$g(x, y) = \begin{cases} f(x, y), & \text{if } (x, y) \in A, \\ 0, & \text{if } (x, y) \in B \setminus A. \end{cases}$$

We define
$$\iint_A f(x, y) dx dy = \iint_B g(x, y) dx dy.$$

Double Integrals

Fubini's Theorem

If $A = [a, b] \times [c, d]$ and $f \in C(A)$ then

$$\iint_A f(x, y) \, dx dy = \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx = \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy.$$

Double Integrals

Fubini's Theorem

If $A = [a, b] \times [c, d]$ and $f \in C(A)$ then

$$\iint_A f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy.$$

Change of Variable

Assume that the mapping $x = X(u, v)$, $y = Y(u, v)$ from the domain T in the uv -plane to the domain S in the xy -plane is bijective, the functions X and Y are continuous and have continuous first order partial derivatives and the Jacobian $J(u, v)$ is never zero. Then

$$\iint_S f(x, y) dx dy = \iint_T f(X(u, v), Y(u, v)) |J(u, v)| du dv.$$

Jointly Distributed Random Variables.

Jointly Distributed Random Variables.

Definition of joint CDF of 2 RVs

Let $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ be two random variables. The function

$$F_{(X,Y)}(x,y) = \mathbb{P}(X \leq x, Y \leq y), \quad (x,y) \in \mathbb{R}^2$$

is called **joint cumulative probability distribution function** (CDF) of X and Y

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.
3. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are right continuous functions.

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.
3. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are right continuous functions.
4. $F_{(X,Y)}(x, +\infty) = F_X(x)$, $x \in \mathbb{R}$, where F_X is the CDF of X . Similarly, $F_{(X,Y)}(+\infty, y) = F_Y(y)$ for any $y \in \mathbb{R}$.

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.
3. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are right continuous functions.
4. $F_{(X,Y)}(x, +\infty) = F_X(x)$, $x \in \mathbb{R}$, where F_X is the CDF of X . Similarly, $F_{(X,Y)}(+\infty, y) = F_Y(y)$ for any $y \in \mathbb{R}$.
5. $F_{(X,Y)}(+\infty, +\infty) = 1$,

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.
3. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are right continuous functions.
4. $F_{(X,Y)}(x, +\infty) = F_X(x)$, $x \in \mathbb{R}$, where F_X is the CDF of X . Similarly, $F_{(X,Y)}(+\infty, y) = F_Y(y)$ for any $y \in \mathbb{R}$.
5. $F_{(X,Y)}(+\infty, +\infty) = 1$,
6. $F_{(X,Y)}(x, -\infty) = F_{(X,Y)}(-\infty, y) = 0$ for any $x, y \in \mathbb{R}$.

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.
3. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are right continuous functions.
4. $F_{(X,Y)}(x, +\infty) = F_X(x)$, $x \in \mathbb{R}$, where F_X is the CDF of X . Similarly, $F_{(X,Y)}(+\infty, y) = F_Y(y)$ for any $y \in \mathbb{R}$.
5. $F_{(X,Y)}(+\infty, +\infty) = 1$,
6. $F_{(X,Y)}(x, -\infty) = F_{(X,Y)}(-\infty, y) = 0$ for any $x, y \in \mathbb{R}$.

Remark: The distribution functions F_X and F_Y are sometimes referred to as the **marginal distribution functions** of X and Y respectively.

Properties of Joint CDF

1. $0 \leq F_{(X,Y)}(x, y) \leq 1$ for any $(x, y) \in \mathbb{R}^2$.
2. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are increasing functions.
3. $F_{(X,Y)}(x, \bullet)$ and $F_{(X,Y)}(\bullet, y)$ are right continuous functions.
4. $F_{(X,Y)}(x, +\infty) = F_X(x)$, $x \in \mathbb{R}$, where F_X is the CDF of X . Similarly, $F_{(X,Y)}(+\infty, y) = F_Y(y)$ for any $y \in \mathbb{R}$.
5. $F_{(X,Y)}(+\infty, +\infty) = 1$,
6. $F_{(X,Y)}(x, -\infty) = F_{(X,Y)}(-\infty, y) = 0$ for any $x, y \in \mathbb{R}$.

Remark: The distribution functions F_X and F_Y are sometimes referred to as the **marginal distribution functions** of X and Y respectively.

Theorem

$$P(a < X \leq b, c < Y \leq d) = \\ F_{(X,Y)}(a, c) + F_{(X,Y)}(b, d) - F_{(X,Y)}(a, d) - F_{(X,Y)}(b, c).$$

Joint PMF of 2 Discrete RVs

In the case when X and Y are both discrete random variables defined on the same sample space, it is convenient to define the **joint probability mass function** (joint PMF) of X and Y by the formula

$$p_{(X,Y)}(x,y) = \mathbb{P}(X = x, Y = y).$$

Joint PMF of 2 Discrete RVs

In the case when X and Y are both discrete random variables defined on the same sample space, it is convenient to define the **joint probability mass function** (joint PMF) of X and Y by the formula

$$p_{(X,Y)}(x, y) = \mathbb{P}(X = x, Y = y).$$

The probability mass function of X can be obtained from the joint PMF $p_{(X,Y)}(x, y)$ by

$$p_X(x) = \sum_{y: p_{(X,Y)}(x,y) > 0} p_{(X,Y)}(x, y)$$

Joint PMF of 2 Discrete RVs

In the case when X and Y are both discrete random variables defined on the same sample space, it is convenient to define the **joint probability mass function** (joint PMF) of X and Y by the formula

$$p_{(X,Y)}(x, y) = \mathbb{P}(X = x, Y = y).$$

The probability mass function of X can be obtained from the joint PMF $p_{(X,Y)}(x, y)$ by

$$p_X(x) = \sum_{y: p_{(X,Y)}(x,y) > 0} p_{(X,Y)}(x, y)$$

Similarly,

$$p_Y(y) = \sum_{x: p_{(X,Y)}(x,y) > 0} p_{(X,Y)}(x, y)$$

Joint PMF of 2 Discrete RVs

In the case when X and Y are both discrete random variables defined on the same sample space, it is convenient to define the **joint probability mass function** (joint PMF) of X and Y by the formula

$$p_{(X,Y)}(x, y) = \mathbb{P}(X = x, Y = y).$$

The probability mass function of X can be obtained from the joint PMF $p_{(X,Y)}(x, y)$ by

$$p_X(x) = \sum_{y: p_{(X,Y)}(x,y) > 0} p_{(X,Y)}(x, y)$$

Similarly,

$$p_Y(y) = \sum_{x: p_{(X,Y)}(x,y) > 0} p_{(X,Y)}(x, y)$$

Remark: The functions p_X and p_Y are sometimes called the marginal PMFs of X and Y respectively.

Jointly Distributed Random Variables.

Example

We are rolling a fair six-sided die. Define two random variables on $\Omega = \{1, 2, 3, 4, 5, 6\}$ as

$$X(\omega) = \begin{cases} 0, & \text{if } \omega = 1, \\ 1, & \text{if } \omega \text{ is composite} \\ 2, & \text{if } \omega \text{ is prime.} \end{cases}$$

$$Y(\omega) = \begin{cases} -5, & \text{if } \omega \leq 3, \\ 10, & \text{if } \omega \geq 4 \end{cases}$$

Construct the joint PMF of X and Y and compute $F_{(X,Y)}(\sqrt{2}, 7)$ and $\mathbb{P}(Y > 5X)$.

Jointly Distributed Random Variables.

Definition

Two random variables X and Y , defined on the same sample space, are called **jointly (absolutely) continuous** if there exists a non-negative function f , defined on the plane, such that the joint CDF of X and Y is representable as

$$F_{(X,Y)}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f(u,v) du dv, \quad (x,y) \in \mathbb{R}^2.$$

The function f is called the **joint probability density function (PDF)** of X and Y .

Computing Marginal PDFs from the Joint PDF

Theorem

If X and Y are jointly continuous random variables then they are also individually continuous. Moreover, their (marginal) PDFs, f_X and f_Y , can be computed from the joint PDF, $f_{(X,Y)}$, as follows:

- $$f_X(u) = \int_{-\infty}^{+\infty} f(u, v) dv$$

- $$f_Y(v) = \int_{-\infty}^{+\infty} f(u, v) du$$

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

a. Find the constant α ;

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

- Find the constant α ;
- Find the marginal density functions (PDFs);

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

- Find the constant α ;
- Find the marginal density functions (PDFs);
- Find the joint and the marginal distribution functions (CDFs);

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

- Find the constant α ;
- Find the marginal density functions (PDFs);
- Find the joint and the marginal distribution functions (CDFs);
- Compute the probability $\mathbb{P}(0 < X < 1; 1 < Y < 2)$;

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

- Find the constant α ;
- Find the marginal density functions (PDFs);
- Find the joint and the marginal distribution functions (CDFs);
- Compute the probability $\mathbb{P}(0 < X < 1; 1 < Y < 2)$;
- Compute $\mathbb{P}(0 < Y/X < 2)$;

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

- Find the constant α ;
- Find the marginal density functions (PDFs);
- Find the joint and the marginal distribution functions (CDFs);
- Compute the probability $\mathbb{P}(0 < X < 1; 1 < Y < 2)$;
- Compute $\mathbb{P}(0 < Y/X < 2)$;
- Find the PDF of the new random variable $Z = Y/X$;

Example

Assume the piecewise defined function

$$f(x, y) = \begin{cases} \alpha \cdot xy \cdot e^{-\frac{x^2+y^2}{2}}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a joint PDF of some random vector (X, Y) .

- Find the constant α ;
- Find the marginal density functions (PDFs);
- Find the joint and the marginal distribution functions (CDFs);
- Compute the probability $\mathbb{P}(0 < X < 1; 1 < Y < 2)$;
- Compute $\mathbb{P}(0 < Y/X < 2)$;
- Find the PDF of the new random variable $Z = Y/X$;
- Calculate $\mathbb{P}(X^2 + Y^2 \leq 4)$.

Signal Processing